SylvaNexus Forest Operations Safety Platform

Real-time multi-hazard early-warning system for mountain forestry

Based on Rahmawati, Yovi & Setiawan (2025) GIS-AHP framework · Live deployment in Taiwan (Baxianshan National Forest) · GitHub Repository

About This Project

SylvaNexus is a real-time occupational safety platform for forest operations in mountainous terrain. It was originally built as a landslide-only warning system using CWA (Taiwan Central Weather Administration) rainfall data and DEM-derived terrain factors.

We have now integrated the **12-parameter GIS-AHP framework** from Rahmawati, Yovi & Setiawan (2025) to upgrade from single-hazard alerts to a **full multi-hazard occupational safety tool**. Your framework — especially the AHP weighting methodology, wind speed, temperature/humidity (WBGT), soil type, and abnormal tree condition criteria — covers exactly the non-landslide hazards we were missing.

Our goal: Protect forest workers' lives by providing site-specific, multi-hazard safety assessments with clear, actionable notifications — not just landslide alerts, but a comprehensive safety picture for the entire work environment.

Platform at a Glance

4 14 3 90+ 5

Hazards Assessed AHP Criteria Region Profiles Test Cases Safety Classes

System Architecture

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|  Backend Service (FastAPI + Python)                          |
|                                                              |
|  Data Sources:                                               |
|    ├─ CWA real-time rainfall (1h / 3h / 24h)                |
|    ├─ CWA weather stations (wind, temp, humidity)            |
|    ├─ CWA 36h township forecasts                            |
|    ├─ CWA earthquake reports (threshold auto-reduction)      |
|    ├─ DEM terrain: slope, TWI, flow accumulation,            |
|    |  geomorphon, aspect, elevation                          |
|    └─ Sentinel-2: canopy height, NDVI                       |
|                                                              |
|  Assessment Engine:                                          |
|    ├─ Landslide: SWCB effective rainfall (Rt) thresholds     |
|    |  4 levels: safe → watch → yellow_alert → red_alert     |
|    ├─ Stream surge: rainfall intensity + flow accumulation   |
|    ├─ Treefall: wind speed + soil saturation correction      |
|    ├─ Heat stress: WBGT (wet-bulb globe temperature)         |
|    └─ GIS-AHP Composite Safety Index (1-5, 5 classes)        |
|       14 criteria, 4 groups, region-specific AHP weights     |
|                                                              |
|  Notifications:                                              |
|    ├─ LINE Notify (field workers)                            |
|    └─ Email (supervisors, local authorities)                 |
└──────────────────────────────────────────────────────────────┘
                       | httpx terrain auto-fetch
                       ▼
┌──────────────────────────────────────────────────────────────┐
|  GIS Service (FastAPI + PostGIS + WhiteboxTools)             |
|                                                              |
|  Spatial Analysis:                                           |
|    ├─ DEM smoothing, slope, geomorphons (Jasiewicz 2013)     |
|    ├─ TWI calculation (Beven & Kirkby 1979)                  |
|    ├─ Flow accumulation, catchment area                      |
|    └─ Per-grid-cell AHP safety index computation             |
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|                                                           |
|  Outputs:                                                 |
|    ├─ GeoJSON safety index map (5-class color-coded)      |
|    ├─ Terrain profile lookup by coordinates               |
|    └─ AHP weight audit endpoint                           |
|                                                           |
|  Database: PostGIS terrain_risk_grid (30m cells)          |
|    + ahp_static_score / ahp_static_class                  |
|    + ahp_composite_score / ahp_composite_class            |
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How Your Framework Is Implemented

AHP Weight Derivation

We implemented the full Analytic Hierarchy Process with pairwise comparison matrices and Saaty's consistency ratio check. Each region has its own set of matrices at two levels: group-level (4 groups) and criterion-level (within each group).

Weights are derived using the geometric mean approximation for the principal eigenvector. Consistency ratios are checked against the 0.10 threshold — if exceeded, an exception is raised and the system refuses to operate with inconsistent weights.

- `app/weather/ahp.py` — AHP engine (156 lines)
- `app/modules/safety/ahp.py` — Ported copy in GIS service

14 Safety Criteria (4 Groups)

We expanded your 12-parameter framework to 14 criteria, organized into 4 groups:

GROUP	CRITERIA	SCORING BASIS	TYPE
Biophysical	Slope	SWCB slope classes (15°/25°/35°/45°)	Static
	Elevation	500/1000/1500/2500m bands	Static

GROUP	CRITERIA	SCORING BASIS	TYPE
	Soil/Geology	slate, mudstone, colluvium, sandstone	Static
	Vegetation (NDVI)	Sentinel-2 NDVI 0.2/0.4/0.6/0.8	Static
	Accessibility	Walking time to road: 15/30/60/120 min	Static
Hydrological	TWI	Beven & Kirkby (1979): 2/4/6/8	Static
	Flow accumulation	1000/5000/10000/50000	Static
	Geomorphon	Jasiewicz & Stepinski (2013) 10 landform classes	Static
	Aspect	Windward vs leeward (typhoon exposure)	Static
Climate	Rainfall (Rt)	Region-specific: CWA 350mm / JMA 200mm / BMKG 100mm	Dynamic
	Wind speed	OSHA Taiwan: 6/10/13/17 m/s + soil saturation correction	Dynamic
	Heat stress (WBGT)	OSHA Taiwan: 25/28/31/33°C	Dynamic
Threat	Wildlife	hornets, bears, snakes, wild boars	Static
	Abnormal trees	Site survey: none / scattered / widespread	Static

Region-Specific AHP Profiles

We defined three region profiles with different pairwise comparison matrices, reflecting different climate, geology, and ecological conditions:

REGION	RAINFALL AUTHORITY	REFERENCE (MM)	KEY DIFFERENCE
Taiwan	CWA / SWCB	350 mm/24h	Typhoon-driven; steep terrain; hornets prominent
Japan	JMA	200 mm/24h	Volcanic soil; bears; lower rainfall threshold
Indonesia	BMKG	100 mm/24h	Tropical; mudstone/lahar; snakes; lowest threshold

All consistency ratios are below 0.10 (verified by automated tests). Weights are exposed via a public API endpoint for expert review and auditability.

Composite vs Static Safety Index

Following your framework's distinction between terrain-inherent risk and dynamic conditions:

- **Static score** — Terrain-only, excludes weather. Used for long-term planning, site comparison, and safety zoning. Computed from 11 static criteria (biophysical + hydrological + threat groups).
- **Composite score** — Includes real-time weather (rainfall, wind, WBGT). Used for dynamic operational safety assessment. The dynamic share (% contribution from weather) is reported so users can see how much current conditions elevate baseline terrain risk.

> **Design principle:** The composite index is a *planning reference*, not an alert trigger. Official SWCB/JMA rainfall thresholds still drive alert levels. This prevents the AHP score from diluting or softening official warnings. (Verified by test: `test_composite_index_does_not_soften_a_threshold_breach`)

Spatial GIS Integration

From Single-Point to Spatial Risk Maps

Your paper's core contribution is the **spatial** GIS-AHP approach — applying the composite index across a grid, not just at one point. We've implemented this:

- **Per-grid-cell computation:** The GIS service reads `terrain_risk_grid` (30m PostGIS cells with slope, TWI, geomorphon, elevation, NDVI, canopy height) and computes the AHP safety index for each cell.
- **GeoJSON output:** The `/safety-index-map` endpoint returns a FeatureCollection where each feature is a grid cell with `static_score`, `static_class`, and color — ready for frontend map rendering.
- **Real terrain auto-fetch:** The backend can call the GIS service to get actual DEM-derived parameters for any coordinate, replacing manual representative values with spatially accurate data.

API Endpoints

Backend (Multi-Hazard Assessment)

GET `/weather/landslide-risk` LIVE

Multi-hazard assessment: landslide (SWCB Rt), stream surge, treefall (wind+soil), heat stress (WBGT). Returns safety_level, composite_index, per-hazard details. Supports `use_gis_terrain=true` with lon/lat for auto-fetching real DEM data.

GET `/weather/safety-criteria` NEW

AHP weight audit: group weights, global weights, consistency ratios, scoring vocabulary, regional rationale. For expert review and transparency.

GIS Service (Spatial Analysis)

`GET` `/api/v1/{project_id}/disaster/safety-index-map` NEW

Per-grid-cell AHP safety index → GeoJSON FeatureCollection. Supports static (terrain-only) and composite (with weather) modes. Bbox spatial filtering.

`GET` `/api/v1/{project_id}/disaster/terrain-profile` NEW

Query terrain parameters at any coordinate. Returns slope, aspect, geomorphon, TWI, flow accumulation, elevation, NDVI, canopy height from PostGIS grid.

`GET` `/api/v1/{project_id}/disaster/ahp-weights` NEW

Full AHP audit: weights, consistency ratios, group/criteria breakdown, rainfall references, geology scores, wildlife hazards per region.

`POST` `/api/v1/{project_id}/disaster/compute-static-risk` UPGRADED

Dual-track computation: legacy 0-100 risk score + AHP 1-5 five-class safety index. Stores both in PostGIS for backward compatibility.

Notification System

When safety_level reaches **yellow_alert** or above, the system automatically sends notifications via:

- **LINE Notify** — to field workers' phones (most common in Taiwan forestry)
- **Email** — to supervisors and local authorities

Messages include the overall safety level, primary hazards, and the composite safety index as a planning reference. Language is safety-oriented and advisory, not alarmist.

The scheduler runs periodic assessments and avoids repeat notifications at the same level to prevent alert fatigue.

Extensions Beyond the Paper

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Real-time CWA Integration

Live rainfall from Taiwan's weather administration, not simulated data. Earthquake-adjusted thresholds (Rt auto-reduced after seismic events).

Sentinel-2 Vegetation

Canopy height and NDVI from Google Earth Engine, integrated into the vegetation criterion scoring.

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Automated Notifications

LINE + Email alerts triggered by safety level, not just risk score. Composite index included as planning reference.

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Spatial Risk Maps

Per-grid-cell AHP computation across entire forest areas, output as GeoJSON for map rendering. Not just single-point assessment.

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Multi-Region Support

Taiwan, Japan, Indonesia profiles with different AHP weights, rainfall authorities, and ecological hazards.

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Full Auditability

AHP weights, consistency ratios, and scoring vocabulary exposed via API. Every score is traceable to its inputs and weights.

Testing & Verification

We maintain 90+ automated tests covering:

TEST SUITE	TESTS	COVERAGE
AHP mechanics	30	Weight derivation, consistency ratio, classification boundaries
Multi-hazard integration	34	Landslide + wind + heat + stream surge, alert gating separation

TEST SUITE	TESTS	COVERAGE
Notification rendering	13	LINE message, email template, repeat suppression
GIS spatial integration	13	safety-index-map, terrain-profile, ahp-weights endpoints

Key test: `test_composite_index_does_not_soften_a_threshold_breach` — ensures the AHP composite index never dilutes official SWCB/JMA alert triggers.

Collaboration Invitation

Dear Azelia,

Thank you for developing the GIS-AHP forest work safety framework. We have implemented your methodology in a live platform for Taiwan's mountain forestry operations, and believe there is significant potential for collaboration.

What we can offer:

- A working real-world deployment with live CWA weather data and DEM terrain
- Implementation of your 12-parameter framework (expanded to 14 criteria) with full AHP weight auditability
- Spatial GIS integration — per-grid-cell safety index maps via PostGIS
- Multi-region support (Taiwan, Japan, Indonesia) with region-specific AHP weights
- Automated LINE + Email notification system for field workers
- 90+ automated tests ensuring correctness and consistency

What we'd love your input on:

- Validation of our AHP pairwise comparison matrices against your expert judgement
- Moran's I spatial autocorrelation validation (we haven't implemented this yet)
- Scoring calibration — are our 1-5 bands aligned with your intended thresholds?
- Extension to other Pacific countries facing climate change forest risks

- Co-authorship on a paper documenting the real-world implementation

As you mentioned — one day, Pacific countries working together on climate change and forest issues would be wonderful. We share that vision.

Best regards,
The SylvaNexus Team
Taiwan 🇹🇼

Repository Access

The full source code is available on GitHub: github.com/Fuilko/SaaSDocker

Note on access: The repository is currently private. We can:

- Add you as a collaborator (simplest — just share your GitHub username)
- Create a public fork with sensitive configs removed
- Share a Docker image so you can run it locally without source access

Please let us know which option works best for you.

Tech stack: Python 3.11, FastAPI, PostgreSQL/PostGIS, Docker, WhiteboxTools, Google Earth Engine, CWA API, LINE Notify API.

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